🗒️ GoNotes — Dokumentasi Teknis (v1.0)

📌 Deskripsi Singkat

GoNotes adalah aplikasi pencatatan berbasis backend Golang, yang dirancang dengan arsitektur modern, containerized dengan Docker, menggunakan PostgreSQL untuk penyimpanan, Redis untuk session & caching, dan dokumentasi API standar industri.

⚙️ Tech Stack

Komponen	Teknologi
Bahasa	Golang (Go)
Web Framework	`chi` / `mux`
Database	PostgreSQL
Migration	`golang-migrate`
Cache / Session	Redis
Container	Docker + Docker Compose
Env Config	.env + `viper`
Auth	JWT + Refresh Token
Docs	YAKK (API collection)

📁 Struktur Folder (v1.0 Planning)

```
gonotes/
├── cmd/              # Entry point (main.go)
├── internal/
│   ├── handler/       # HTTP layer (Controller)
│   ├── service/       # Usecases
│   ├── repository/     # DB/Redis layer
│   ├── model/         # Struct definitions
│   ├── middleware/     # Auth, logging
```

```
│   ├── config/          # Env loader
│   └── utils/           # Helper functions
├── migrations/          # SQL migration files
├── docs/                # API documentation
├── .env                 # Local environment vars
├── docker-compose.yaml  # Dev setup
├── Dockerfile
└── README.md
```

🧾 Format Standar Response API

```
{
  "status": "success",     // success | error
  "code": 200,             // HTTP code
  "message": "Optional info",
  "data": {},              // Optional data (note, user, etc.)
  "error": null            // Or error message
}
```

📡 Konvensi Status Code

Kode	Keterangan
200	OK
201	Created
204	No Content
400	Bad Request (validasi gagal)
401	Unauthorized (token invalid)
403	Forbidden (akses ditolak)
404	Not Found
409	Conflict
422	Invalid Input Format
500	Internal Server Error

🧩 Struktur Database (v1 MVP)

users

```
CREATE TABLE users (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  email TEXT UNIQUE NOT NULL,
  password TEXT NOT NULL,
  full_name TEXT NOT NULL,
  created_at TIMESTAMP DEFAULT now(),
  updated_at TIMESTAMP DEFAULT now()
);
```

notes

```
CREATE TABLE notes (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  user_id UUID NOT NULL REFERENCES users(id) ON DELETE CASCADE,
  title TEXT NOT NULL,
  content TEXT,
  created_at TIMESTAMP DEFAULT now(),
  updated_at TIMESTAMP DEFAULT now()
);
```

sessions

```
CREATE TABLE sessions (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  user_id UUID NOT NULL REFERENCES users(id),
  refresh_token TEXT NOT NULL,
  user_agent TEXT,
  ip_address TEXT,
  is_valid BOOLEAN DEFAULT TRUE,
  created_at TIMESTAMP DEFAULT now(),
  expires_at TIMESTAMP
);
```

🔐 Redis (Digunakan Untuk)

Tujuan	Keterangan
Session	JWT/Refresh token cache
Caching	Hasil query (notes list, dsb)
Rate Limit	Batasan request API

🔧 .env File Contoh

```
APP_PORT=8080

# PostgreSQL
DB_HOST=db
DB_PORT=5432
DB_USER=postgres
DB_PASSWORD=postgres
DB_NAME=gonotes

# Redis
REDIS_HOST=redis
REDIS_PORT=6379

# Auth
JWT_SECRET=supersecretkey
JWT_EXPIRE=15m
REFRESH_EXPIRE=7d
```

📌 API Caching (Contoh Dokumentasi)

Endpoint	Caching	TTL	Invalidated by
GET /notes	Redis	300s	POST/PUT/DELETE /notes
GET /profile	Redis	180s	PUT /profile

🚧 Roadmap / Rencana Batch Pengembangan

🎯 Tujuan: Buat Per Modul → Tes → Review → Lanjut

✅ Batch 1 — Project Init & Auth Basic

- [ ] Setup struktur folder
- [ ] Setup `.env` + config loader
- [ ] Setup Dockerfile & docker-compose
- [ ] Migration awal: `users`, `sessions`
- [ ] Implementasi Register
- [ ] Implementasi Login (JWT + Redis + DB session)
- [ ] Middleware Auth
- [ ] Refresh Token Endpoint
- [ ] Logout
- [ ] Dokumentasi Auth di YAKK

🔄 Batch 2 — Notes Module

- [ ] CRUD Notes (by user)
- [ ] Pagination + Filtering
- [ ] Redis cache untuk GET notes
- [ ] Auto-invalidate cache setelah perubahan
- [ ] Dokumentasi endpoint Notes di YAKK

🔁 Batch 3 — User Module

- [ ] Get Profile
- [ ] Update Profile
- [ ] Caching profile di Redis
- [ ] Dokumentasi YAKK

🔐 Batch 4 — Session & Security

- [ ] Lihat semua session aktif

- [ ] Logout session tertentu (device)
- [ ] Middleware logging & rate limit
- [ ] Audit log (opsional)

🧪 Batch 5 — Testing & Docs

- [ ] Unit test Auth & Notes
- [ ] Integration test Docker
- [ ] Dokumentasi YAKK lengkap
- [ ] README cara run project